

Semantics-guided 3D Indoor Scene Reconstruction from a Single RGB Image with Implicit Representation

Yi-Ju Pan
National Yang Ming Chiao Tung
University
Taiwan
e.p.880107@gmail.com

Pei-Chun Tsai
National Yang Ming Chiao Tung
University
Taiwan
peichun23.cs12@nycu.edu.tw

Kuan-Wen Chen
Department of Computer Science,
National Yang Ming Chiao Tung
University
Taiwan
kuanwen@cs.nycu.edu.tw

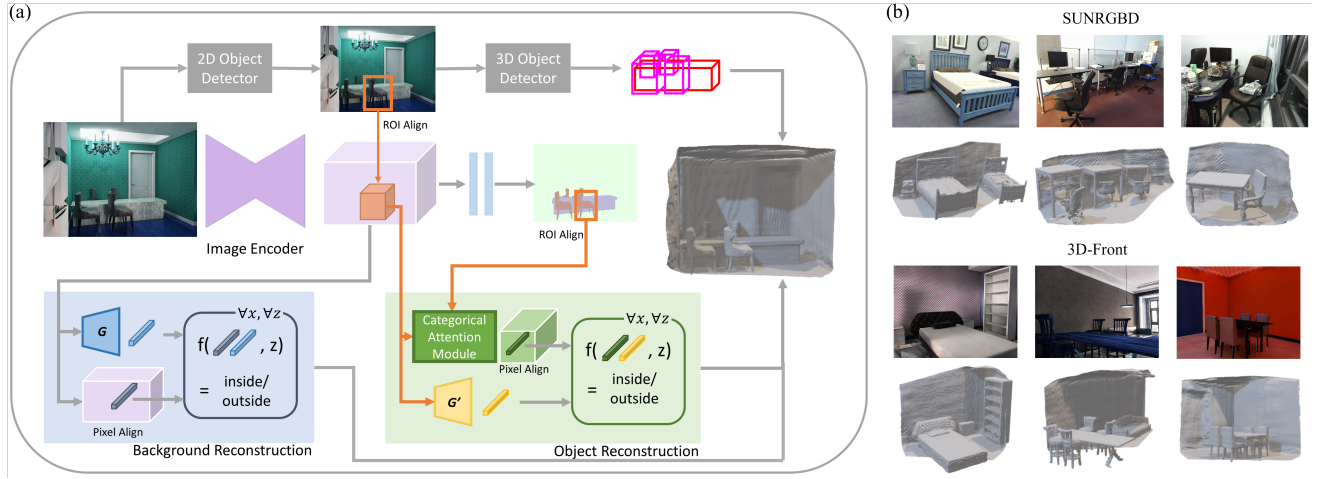


Figure 1: (a) Architecture of proposed SemanticPIFu. (b) Scene reconstruction results for SUNRGBD [Song et al. 2015] and 3D-Front [Fu et al. 2021a].

Abstract

Single-view 3D scene reconstruction is challenging due to limited single-image information. Existing methods often overlook the importance of image feature extraction. We propose a novel approach that integrates semantic segmentation with an implicit function, using a semantics-guided image encoder to enhance feature extraction for improved background and object reconstruction. A categorical attention module based on semantic segmentation further aids object reconstruction. Experimental results show our method's strong performance in both background and object reconstruction.

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1 Introduction

Reconstruction of 3D scenes from a single image is a downstream task in computer vision for applications in robotics and AR/VR. As deep learning techniques have evolved, several methods for directly predicting object shapes have been proposed. For example, Im3D [Zhang et al. 2021] employ implicit surface prediction to achieve fine-grained object reconstruction. Inspired by PIFu [Saito et al. 2019], InstPIFu [Liu et al. 2022] combines instance-aligned pixel features and global features, thereby enhancing detailed predictions and achieving enhanced generality. Furthermore, InstPIFu is the first approach to introduce background reconstruction, replacing previous simplified background representations.

Although existing methods [Liu et al. 2022; Zhang et al. 2021] have produced noteworthy results, they overlook the importance of image feature extraction. We proposed a method that leverages semantic segmentation predictions to directly guide the image encoder and weight pixel features. Our segmentation prediction head helps the encoder capture image features with category information, reducing the number of stacks in the encoder and leading to improved object and background reconstruction with a more compact network. Additionally, our novel categorical attention module effectively directs the network's attention toward the object of interest, thereby enhancing the network training process.

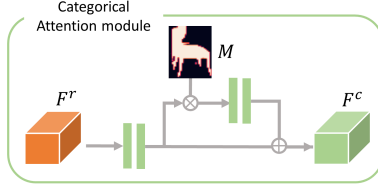


Figure 2: Architecture of categorical attention module. Feature F^r is processed through 2D convolution blocks and the mask M to obtain the weighted feature F^c .

The experimental results show the superior accuracy of our approach on both datasets [Fu et al. 2021a; Sun et al. 2018] compared with InstPIFu [Liu et al. 2022]. Notably, our method is one of the few capable of handling both background and object reconstruction.

2 Method

The proposed SemanticPIFu architecture, shown in Fig. 1, consists of three main components: object detection, background reconstruction, and object reconstruction. The final scene reconstruction is assembled by combining the predictions from these components. Object detection encompasses both 2D and 3D object detection, where the 2D detector predicts object categories and 2D bounding boxes, while the 3D detector extends these predictions to 3D poses.

Background reconstruction comprises two main modules: the proposed semantic-guided image encoder and the adapted PIFu from InstPIFu [Liu et al. 2022]. The semantic-guided image encoder extracts features from the input image, which are then used by the adapted PIFu to predict the background surface. Object reconstruction is completed using the semantic-guided image encoder and the adapted PIFu enhanced with our proposed categorical attention module. After image features are extracted, RoI alignment is performed using a predicted 2D bounding box to acquire instance-related features, which are then processed by the categorical attention module before pixel alignment.

For our semantic-guided image encoder, we added a shared semantic segmentation prediction head of 2D CNNs after each stack to guide the encoder. This integration enables the network to learn features containing category information, reducing the number of stacks needed and yielding better results with a relatively light-weight encoder.

After obtaining image features, we introduce the categorical attention module to enhance the network’s focus on the object of interest before extracting pixel features. We extract semantic segmentation predictions based on the category label from object detection. Through RoI alignment, we obtain predicted probability values within the 2D bounding box, which serves as the mask M . This process acts as a feature weighting mechanism, utilizing the probability mask to direct the network’s attention to high-probability areas, highlighting regions corresponding to the object to be reconstructed. The architecture is depicted in Fig. 2.

Background reconstruction and object reconstruction are trained separately. Both networks use cross-entropy loss for 2D image semantic segmentation and the mean squared error between the predicted occupancy and ground truth values of the 3D points.

Table 1: Comparison of room background reconstruction on 3D-Front [Fu et al. 2021a].

Method	CD ↓
Im3D [Zhang et al. 2021]	1.974
InstPIFu [Liu et al. 2022]	0.481
Ours	0.294

Table 2: Comparison of object reconstruction results on Pix3D [Sun et al. 2018] and 3D-Future [Fu et al. 2021b], measured by mean CD (1e-3) and mean F-score.

Method	Pix3D		3D-Future	
	CD ↓	F-Score ↑	CD ↓	F-Score ↑
LIEN [Zhang et al. 2021]	51.31	31.45	28.52	45.63
InstPIFu [Liu et al. 2022]	24.65	45.62	14.46	61.32
SSR [Chen et al. 2024]	21.79	59.71	10.45	71.36
Ours	18.87	47.19	12.70	68.06

3 Experiments and Discussions

We conducted background reconstruction experiments on 3D-Front [Fu et al. 2021a], with results presented in Table 1. Compared with baseline [Liu et al. 2022], the CD decreased by 0.187.

For object reconstruction, we performed experiments on Pix3D [Sun et al. 2018] and 3D-Future [Fu et al. 2021b], with the quantitative results showed in Table 2. Our method achieved the best CD score on Pix3D, significantly outperforming InstPIFu [Liu et al. 2022]. While slightly behind SSR [Chen et al. 2024] on other metrics, our method can reconstruct both the background and objects, restoring the entire scene, which distinguishes it from their approach.

We evaluated the complete scene reconstruction on SUNRGBD [Song et al. 2015] and 3D-Front [Fu et al. 2021a], with visualization results shown in Fig. 1 (b). Our method achieved the best performance among those capable of reconstructing both background and foreground objects, resulting in high-quality scene reconstruction.

Acknowledgments

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References

- Yixin Chen, Junfeng Ni, Nan Jiang, Yaowei Zhang, Yixin Zhu, and Siyuan Huang. 2024. Single-view 3D Scene Reconstruction with High-fidelity Shape and Texture.
- H. Fu, B. Cai, L. Gao, L.-X. Zhang, J. Wang, C. Li, Q. Zeng, C. Sun, R. Jia, B. Zhao, et al. 2021a. 3d-front: 3d furnished rooms with layouts and semantics. In *ICCV*. 10933–10942.
- H. Fu, R. Jia, L. Gao, M. Gong, B. Zhao, S. Maybank, and D. Tao. 2021b. 3d-future: 3d furniture shape with texture. *International Journal of Computer Vision* (2021), 1–25.
- H. Liu, Y. Zheng, G. Chen, S. Cui, and X. Han. 2022. Towards High-Fidelity Single-view Holistic Reconstruction of Indoor Scenes. In *ECCV*.
- S. Saito, Z. Huang, R. Natsume, S. Morishima, A. Kanazawa, and H. Li. 2019. PIFu: Pixel-Aligned Implicit Function for High-Resolution Clothed Human Digitization. In *ICCV*.
- S. Song, S. P. Lichtenberg, and J. Xiao. 2015. SUN RGB-D: A RGB-D scene understanding benchmark suite. In *CVPR*.
- X. Sun, J. Wu, X. Zhang, Z. Zhang, C. Zhang, T. Xue, J. B. Tenenbaum, and W. T. Freeman. 2018. Pix3D: Dataset and Methods for Single-Image 3D Shape Modeling. In *CVPR*.
- C. Zhang, Z. Cui, Y. Zhang, B. Zeng, M. Pollefeys, and S. Liu. 2021. Holistic 3D Scene Understanding From a Single Image With Implicit Representation. In *CVPR*. 8833–8842.